Feature Breakdown Document

Project: Web-Based Educational Platform for Centralized Academic Resources **Document type:** Feature Breakdown **Version:** 1.0

1. Overview

This document breaks the platform into modules, features and user stories, with priorities, acceptance criteria and release phases. It is meant to be the working checklist for development and testing.

Priority key: P1 = Must have (Phase 1 MVP), P2 = Should have (Phase 2), P3 = Could have / future (Phase 3 and beyond).

2. Feature Map

Module	Features
A. Student Portal	Home, content browsing, resource viewing and download, lab section, question papers, announcements, feedback, contact, search, filters
B. Admin Dashboard	Authentication, structure management, resource management, announcements, feedback inbox, dashboard overview, settings
C. Platform Services	File handling, validation, security, notifications, backups
D. Future Enhancements	Advanced search, video lectures, discussion forums, AI assistant, student accounts

3. Module A: Student Portal

A1. Home Page (P1)

- Hero section with platform name and short description
- Quick links to semesters, lab resources, question papers

- Latest announcements panel
- Recently added resources list
- **Acceptance:** Home page loads under 3 seconds and links reach every main section.

A2. Hierarchical Content Browsing (P1)

- Semester list → Subject list → Module list → Resource list
- Breadcrumb navigation and back links
- Subject cards showing code, name and resource count
- **User story:** As a student, I want to open my semester and subject so that I can find my syllabus modules quickly.
- **Acceptance:** Any resource is reachable within 4 clicks from the home page.

A3. Resource Viewing and Download (P1)

- In-browser PDF viewer
- Download button for PDF, DOCX, PPTX, ZIP and code files
- Metadata display: title, type, upload date, file size
- **User story:** As a student, I want to preview a file before downloading so that I do not waste data on the wrong document.
- **Acceptance:** PDFs open in the viewer; all supported formats download with the correct file name.

A4. Laboratory Resources (P1)

- Lab section organized by subject
- Lab manuals, experiment lists, sample programs and output screenshots
- Source code files downloadable individually or as a ZIP
- **Acceptance:** Each lab subject shows its manual and programs in a clear list.

A5. Previous Year Question Papers (P1)

- Papers grouped by subject and exam year
- Filter by year and exam type (internal, university)
- **Acceptance:** A student can open the latest paper for a subject in at most 3 clicks.

A6. Announcements (P1)

- Announcement list on the home page and a dedicated page
- New items highlighted; expired items hidden automatically
- **Acceptance:** Published announcements appear immediately and expired ones disappear.

A7. Feedback Form (P1)

- Fields: name, email, category (content, bug, suggestion, other), message, optional rating
- Validation and success confirmation
- Spam protection (CAPTCHA or rate limit)
- **Acceptance:** Valid submissions are stored and visible to the admin; invalid ones show clear error messages.

A8. Contact Administrator (P1)

- Contact form with name, email, subject and message
- Optional display of admin contact email
- **Acceptance:** Submitted messages appear in the admin inbox.

A9. Basic Search (P2)

- Search box in the header
- Matches title, subject name, subject code and keywords
- Results grouped by subject with type labels
- **Acceptance:** Searching a subject code returns its resources first.

A10. Filters and Sorting (P2)

- Filter by resource type, year and format
- Sort by newest, title or most downloaded

A11. Responsive and Accessible Design (P1)

- Mobile-first layout, large tap targets
- Readable contrast, alt text and keyboard navigation
- Light and dark mode (P3)

A12. Static Pages (P2)

- About, help or FAQ, privacy note, terms of use and copyright or takedown notice

4. Module B: Admin Dashboard

B1. Secure Authentication (P1)

- Login and logout, password hashing, session timeout
- Lockout or delay after repeated failed attempts
- Password change (P2) and password reset by email (P3)
- **Acceptance:** Any /admin page redirects to login when unauthenticated.

B2. Academic Structure Management (P1)

- Add, edit, delete and reorder semesters, subjects and modules
- Warning before deleting items that contain resources
- **User story:** As an admin, I want to create a new subject and its modules so that resources can be uploaded under it.

B3. Resource Management (P1)

- Upload form: title, description, semester, subject, module, type, year, file
- Edit metadata, replace file, delete
- Publish or unpublish toggle (P2)
- Multiple file upload (P2) and bulk upload via ZIP or CSV mapping (P3)
- **Acceptance:** A valid upload is visible on the student side immediately after publishing; invalid files are rejected with a clear message.

B4. Announcement Management (P1)

- Create, edit, delete and set expiry date
- Pin an important announcement (P2)

B5. Feedback and Message Inbox (P1)

- List with unread indicator, view details, mark read, delete
- Filter by category and date (P2)

- Export to CSV (P3)

B6. Dashboard Overview (P2)

- Counts: total resources, subjects, unread messages
- Recent uploads and recent feedback
- Download statistics per resource (P3)

B7. Admin Accounts and Roles (P3)

- Multiple admins with roles: super admin and content admin
- Activity log of uploads and deletions

5. Module C: Platform Services

ID	Feature	Priority	Notes
C1	File validation (type, size, safe naming)	P1	Allow-list of formats, size cap such as 25 MB
C2	Database persistence and migrations	P1	Versioned schema changes
C3	Input sanitization, CSRF and XSS protection	P1	Applied to all forms
C4	HTTPS and secure configuration	P1	Production requirement
C5	Error pages (404, 500) and logging	P1	Friendly messages for users
C6	Automated backups	P2	Daily database, weekly files
C7	Email notifications to admin	P3	On new feedback or message
C8	Download counter	P3	Increment on each download
C9	Basic analytics	P3	Page views and popular resources

6. Module D: Future Enhancements

ID	Feature	Description	Dependencies
D1	Advanced search	Full-text search inside documents, typo tolerance, suggestions	PDF text extraction, search engine
D2	Video lectures	Video entries per module, embedded or hosted	Video hosting or object storage
D3	Discussion forums	Subject-wise threads, replies, moderation	Student accounts, moderation tools
D4	AI academic assistant	Answer questions and summarize notes using the platform's resources	Indexed content, LLM API, usage limits
D5	Student accounts	Bookmarks, recent activity, personalized dashboard	Authentication expansion
D6	Ratings and comments on resources	Students rate material quality	Student accounts
D7	Mobile app or PWA	Offline access to downloaded material	Stable API
D8	Syllabus and exam timetable pages	Central academic reference	Content from institution

7. Release Scope Summary

Release	Included features
MVP (Phase 1)	A1–A8, A11, B1–B5, C1–C5
Phase 2	A9, A10, A12, B6, publish toggle, multi-upload, C6, password change
Phase 3	B7, C7–C9, bulk upload, dark mode, CSV export
Future	D1–D8

8. Feature Dependencies

- Resource upload (B3) depends on academic structure (B2).
- Student browsing (A2–A5) depends on published resources from B3.
- Search (A9) depends on stable metadata fields in the Resource entity.
- Forums (D3) and bookmarks (D5) depend on student accounts.
- The AI assistant (D4) depends on advanced search or an indexed content store (D1).

9. Non-Functional Features Checklist

Area	Target
Performance	Pages under 3 seconds, paginated lists
Security	Hashed passwords, validated uploads, HTTPS
Compatibility	Latest Chrome, Firefox, Edge and Safari; Android and iOS browsers
Accessibility	WCAG 2.1 AA basics
Maintainability	Modular code, README, commented critical logic

10. Testing Focus per Module

Module	Key tests
Student portal	Navigation paths, PDF view, downloads of each format, form validation, mobile layout
Admin dashboard	Login protection, CRUD on structure and resources, oversize and wrong-type upload rejection
Platform services	SQL injection and XSS attempts, CSRF checks, backup restore test
Integration	Upload as admin then find and download as student; announcement lifecycle

11. Definition of Done (per feature)

- Meets acceptance criteria listed above
- Code reviewed and merged to main branch
- Unit and basic integration tests pass
- Works on desktop and mobile
- No open critical or high-severity bugs
- Documented in the README or user guide where relevant